

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	06 July 2026
Team ID	[Enter Your Team ID]
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Technical Architecture:

The deliverable below shows the architectural diagram for the Heritage Treasures UNESCO Tableau project, along with the information in Table 1 and Table 2.

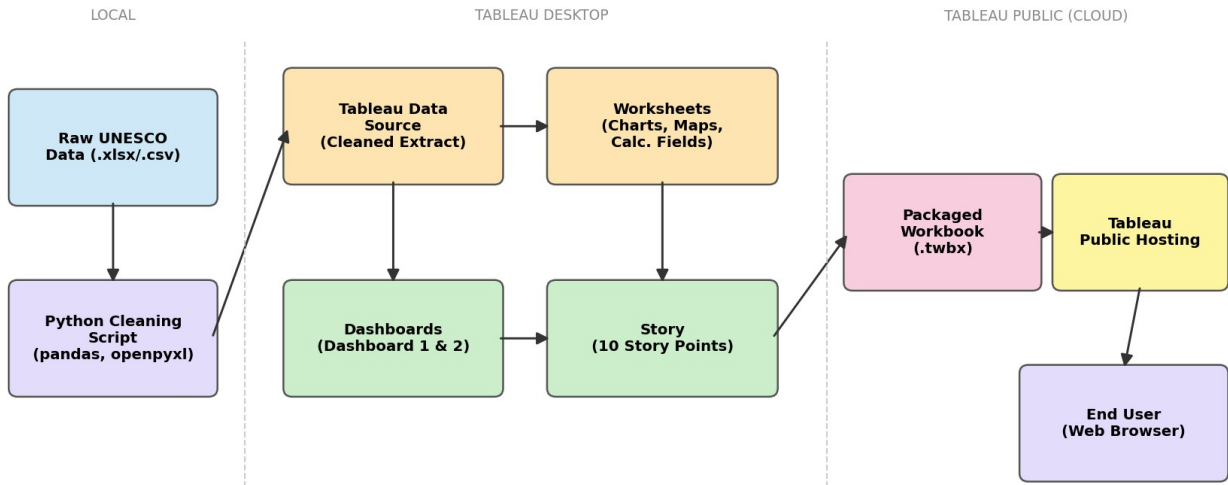


Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the viewer interacts with the dashboard	Tableau Public web viewer (browser-based, HTML5/JS rendering)
2.	Data Source	Raw UNESCO World Heritage Sites dataset	Excel (.xlsx) — 1,121 rows × 18 columns
3.	Data Cleaning / ETL	Removing HTML entities, whitespace, and duplicates; validating fields	Python (pandas, openpyxl)
4.	Data Modeling	Structuring cleaned data for visualization	Excel workbook: UNESCO_Clean_Data and Summary sheets
5.	Visualization Engine	Building charts, dashboards, and the guided story	Tableau Desktop

6.	Publishing / Hosting	Hosting the workbook for public access	Tableau Public (cloud hosting)
7.	File Storage	Storage of source Excel file and Tableau workbook	Local filesystem during development; Tableau Public cloud storage after publishing
8.	External API	Map rendering service for the world map view	Mapbox / OpenStreetMap (built into Tableau)
9.	Machine Learning / Forecast Model	Forecasting future heritage site inscriptions	Tableau built-in forecasting (exponential smoothing) on the Year Forecasting sheet
10.	Infrastructure	Application development and deployment environment	Local machine (Tableau Desktop) for development; Tableau Public cloud for deployment

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Libraries used for data cleaning and validation	Python: pandas, openpyxl
2.	Security Implementations	Data handling and access controls	Only public, non-personal UNESCO open data used; access limited to a read-only public URL
3.	Scalable Architecture	Justification for scalability of the workbook design	Tableau Data Extract (.hyper) refreshed as new UNESCO sites are inscribed each year
4.	Availability	Justification for application availability	Hosted on Tableau Public's cloud infrastructure, available independent of the author's device
5.	Performance	Design considerations for dashboard performance	Tableau Hyper in-memory engine and pre-aggregated calculated fields for fast filter response

References:

- <https://public.tableau.com/>
- <https://help.tableau.com/current/pro/desktop/en-us/>
- <https://whc.unesco.org/en/list/>
- <https://c4model.com/>